

Date	May 11, 2026
Reporter	James Neil M. Maano
Project	TrioTagaPayo AMS DBTC
Report Period	March 7, 2026 – May 11, 2026
Sprint / Period	Current Sprint

STAND-UP REPORT

What was achieved this period?

Previously completed (March 7 – April 25, 2026):

- Set up Backend folder/project structure and pushed to GitHub
- Implemented QR Code Attendance System (QRSession, QRScan, IQRService, QRSessionRepository)
- Integrated EmailJS Email Notification System with failure isolation
- Enhanced Attendance Repository with filter and bulk interfaces
- Added Global Exception Middleware with structured JSON error responses
- Configured Program.cs with all repositories, services, JWT, Swagger, CORS, admin seeding
- Implemented API Rate Limiting (login, global, qrscan policies)
- Implemented Refresh Token with Rotation (RefreshTokenHelper, POST /api/Auth/refresh)
- Implemented SignalR Realtime Notifications (AttendanceHub, NotifyHubAsync)
- Implemented JWT Cookie Session (HttpOnly cookies, logout, GET /api/Auth/me, ClaimsHelper)

May 5, 2026 achievements (already in previous report):

- Implemented Admin and Teacher Role-Based Account Hierarchy
- Added TeacherId FK to User model with proper DB level link
- Added teacher_id claim inside JWT and GetTeacherId() helper
- Added ChangePassword endpoint for all roles
- Blocked disabled teacher login and token refresh
- Fixed TeacherService to use TeacherId FK lookup
- Resolved Render deployment issues (connection string, environment variables, EmailJS template)

NEW – May 9 – May 11, 2026:

Email Verification for Students

- Added IsEmailVerified, EmailVerificationToken, EmailVerificationTokenExpiry to User model
- Created POST /api/Auth/register — now accepts real email address (updated RegisterDto)
- Created POST /api/Auth/verify-email endpoint to confirm token
- Created POST /api/Auth/resend-verification endpoint
- Integrated SMTP email service (using MailKit + Gmail App Password) because free EmailJS cannot send to external domains
- Added EmailSettings to appsettings.json (SmtpServer, Port, SenderEmail) and stored password in user secrets / Render env variables

Admin Password Reset (Forgot Password Scenario)

- Added POST /api/Admin/reset-user-password endpoint (Admin only)
- Admin can reset any user's password without old password
- Added AdminResetPasswordDto and integrated with IUserRepository
- (Optional) Added POST /api/Auth/forgot-password-request – notifies admin, no automatic reset

Swagger Documentation Improvements

- Added missing [ProducesResponseType] attributes to all endpoints in AccountController, AuthController, EnrollmentController, etc.
- Enabled XML documentation (GenerateDocumentationFile) so parameter descriptions appear in Swagger
- Added clear /// <param name="..."> comments for query parameters (page, pageSize, etc.)

Cleaner API Structure

- Removed duplicate CRUD methods from AdminController (students, teachers, courses, attendance now only in their dedicated controllers)
- AdminController now only holds dashboard and reset-user-password
- Dedicated controllers (StudentController, TeacherController, CourseController, AttendanceController) have proper [Authorize(Roles = "...")]

SMTP Email Service Implementation

- Created SmtpEmailService.cs using MailKit
- Used HTML email body with verification link
- Graceful failure – email errors do not break registration, only logged

Production (Render) Configuration

- Added EmailSettings__SenderPassword as environment variable (App Password)
- Verified that SMTP can send to any external email (Gmail, Yahoo, school domain)

What is the focus today (May 11)?

- Final testing of SMTP email verification on live Render deployment
- Coordinate with frontend team to consume the new /verify-email endpoint
- Document all API endpoints (Swagger) and share with frontend
- Ensure email verification is required before login (already implemented in AuthService.LoginAsync)
- Test admin password reset flow end to end

What blockers exist?

- No major blockers now – SMTP works after setting App Password.
- Potential future issue: Render free tier may block port 587? Currently working.
- Reminder: EmailJS still used for attendance notifications (works fine); SMTP only for verification.

RETROSPECTIVE

What went well? (Wins/Successes)

- Email verification was implemented cleanly without breaking existing login flow
- SMTP approach gives full control over email delivery – no third party API limits (except Gmail daily sending quota)
- Swagger documentation is now complete – frontend team can see all possible status codes and parameter descriptions
- Admin reset password endpoint allows school staff to help students who forget their passwords
- Codebase is now well organised – each resource has its own controller, no duplication

What didn't go well? (Problems/Challenges)

- Free EmailJS cannot send to external emails – discovered late, forced SMTP integration
- Gmail App Password generation required extra steps and Google 2 Factor Authentication
- Initial SMTP connection failed because port 587 needed useSsl = false (STARTTLS) – corrected after debugging
- Render environment variable naming required double underscore (EmailSettings__SenderPassword) not single dot

What should we start doing? (Improvements)

- Write integration tests for email verification flow (to catch failures early)
- Add a fallback email provider (SendGrid) in case Gmail rate limits
- Use a dedicated email sending service for production (e.g., Resend, SendGrid) instead of Gmail SMTP for higher quota

- Log all email errors to a file or monitoring service

What should we stop doing? (Waste/Inefficiencies)

- Stop relying on free EmailJS for critical verification emails (keep it only for notifications)
- Stop hardcoding email configurations – always use environment variables

What should we continue doing? (Best Practices)

- Maintain conventional commit format (feat, fix, chore, docs)
- Keep helper classes separated and well documented
- Apply [ProducesResponseType] on all endpoints
- Test with real external email addresses before marking feature complete

Next Steps

- Merge all changes to main branch and tag release v1.1.0
- Deploy latest backend to Render (already done)
- Help frontend team integrate the /verify-email page
- Monitor SMTP email delivery logs
- Prepare for student enrollment with real email verification

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